🌈 Depth Pseudo Color Image Rendering (Depth to Color)

1. Course Content

By subscribing to the raw single-channel depth data from the Angstrong Nuwa-HP60C depth camera (`/ascamera_hp60c/camera_publisher/depth0/image_raw`), use OpenCV's colormap algorithm to perform pseudo-color rendering on the continuous depth space, generating a heatmap that intuitively reflects the depth of objects.

> **Important**
>
> This course is only applicable to users with the **Angstrong Nuwa-HP60C Depth Camera** version.

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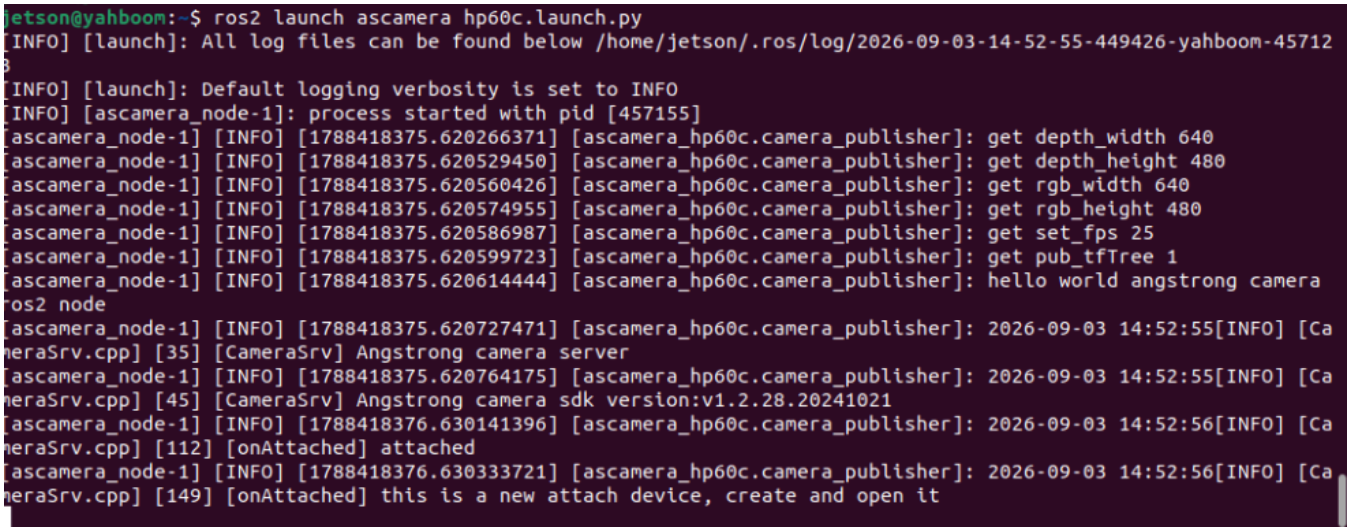
2. Preparation and Running

This case requires opening 2 terminals to start the camera driver and pseudo-color rendering node respectively:

2.1 Start Camera Low-level Driver

Run in **Terminal 1**:

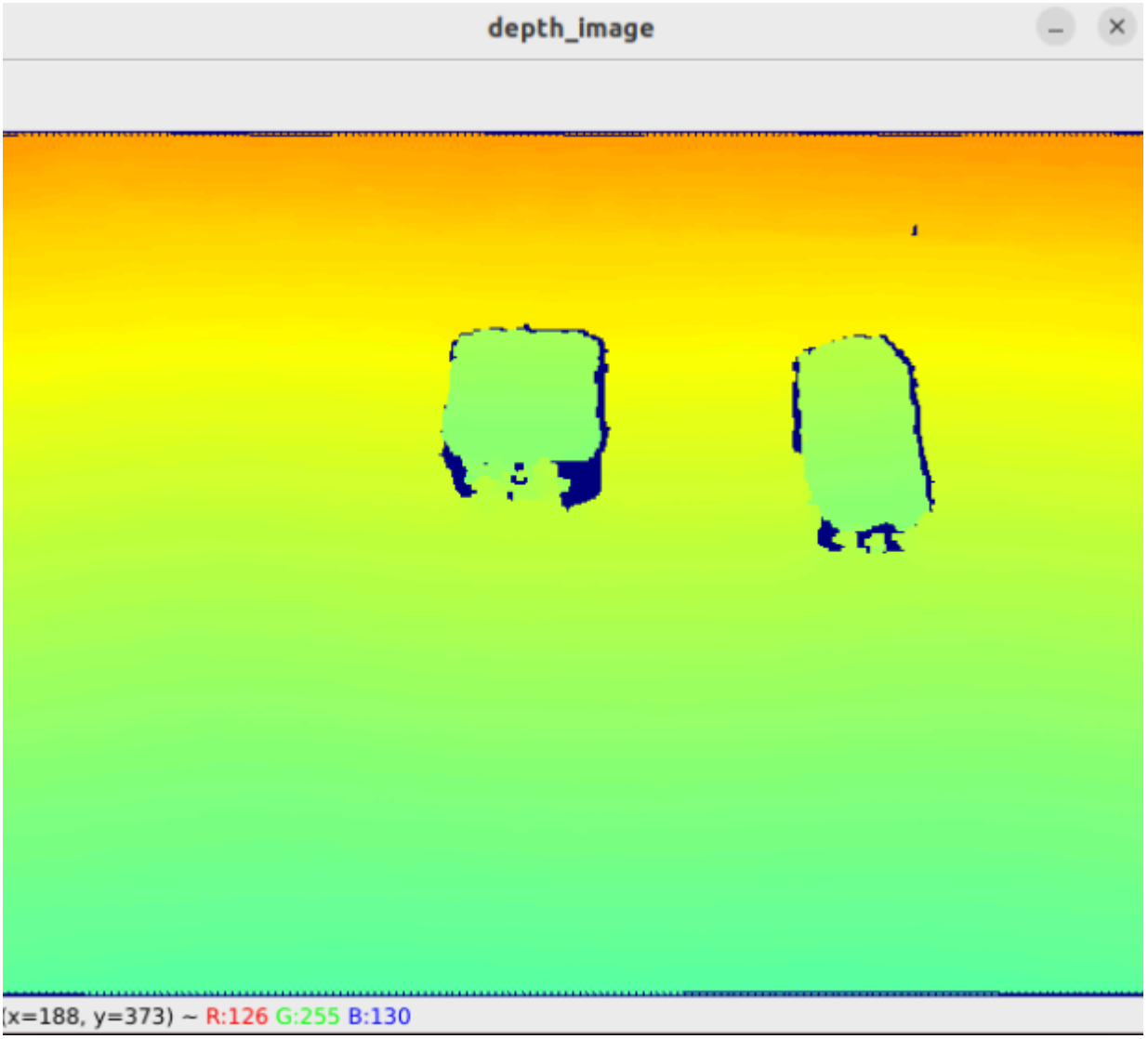
```
ros2 launch ascamera hp60c.launch.py
```



2.2 Start Pseudo Color Image Rendering Node

Run in **Terminal 2**:

```
ros2 run yahboomcar_depth depth_to_color
```



3. Core Source Code Analysis

Source Code Path:

```
$HOME/yahboomcar_ros2_ws/yahboomcar_ws/src/yahboomcar_depth/yahboomcar_depth/Basic/depth_to_color.py
```

```
class DepthToColor(Node):
    def __init__(self):
        super().__init__('depth_to_color')
        # 1. Subscribe to Nuwa depth image topic (16UC1 / 32FC1 millimeter unit)
        self.sub = self.create_subscription(
            Image, '/ascamera_hp60c/camera_publisher/depth0/image_raw', self.topic, 10)
        self.window_name = 'depth_image'
        self.depth_bridge = CvBridge()

    def topic(self, msg):
        # 2. Convert ROS depth image to 32-bit single-precision floating point matrix
        depth_image_orin = self.depth_bridge.imgmsg_to_cv2(msg, '32FC1')
```

```
        # 3. Linear grayscale stretch and convert to 8-bit single-channel image (alpha
adjusts contrast scaling)
        depth_scaled = cv.convertScaleAbs(depth_image_orin, alpha=0.45)

        # 4. Apply JET classic scientific pseudo-color mapping (red for near, blue for far)
        depth_to_color_image = cv.applyColorMap(depth_scaled, cv.COLORMAP_JET)

        # 5. Render and display
        cv.imshow(self.window_name, depth_to_color_image)
        cv.waitKey(1)
```

3.1 OpenCV Common Colormap Description

`cv.applyColorMap` supports multiple classic pseudo-color palettes:

Colormap Macro	Color Gradient Style	Applicable Scenario
`cv.COLORMAP_JET`	Blue → Cyan → Yellow → Red (classic heatmap)	3D depth camera depth visualization (default)
`cv.COLORMAP_VIRIDIS`	Blue-purple → Green → Yellow (modern perceptually uniform colormap)	Scientific visualization and high contrast imaging
`cv.COLORMAP_HOT`	Black → Red → Yellow → White	Thermal imaging and high dynamic temperature mapping
`cv.COLORMAP_BONE`	Black → Blue-gray → White	Similar to X-ray grayscale medical style
`cv.COLORMAP_RAINBOW`	Red → Orange → Yellow → Green → Blue → Purple	Rainbow color multi-band high contrast differentiation